Group 2 – BSIT 3-5 | Intro to Data Science | CRISP-DM

Main Idea

Build a time-series forecasting model that predicts retail prices of key food commodities (rice, vegetables, fish) several weeks ahead. The study also analyzes which shocks (typhoons, fuel prices, imports) are most associated with price spikes. The focus is well-milled rice in Metro Manila, forecast 4 weeks ahead.

The Problem

- Sudden price spikes in staple foods hit low-income households hardest, since food takes up a large share of their budget.
- The DA and traders need advance warning to act on spikes, but prices are currently followed reactively.
- It is unclear which shocks (typhoons, fuel costs, import volumes) actually drive the spikes.

Business Question

Can we forecast the retail price of well-milled rice in Metro Manila 4 weeks ahead well enough to give the DA and traders an early warning of price spikes, and which shocks (typhoons, fuel prices, import volumes) are most associated with those spikes?

Type	**Question**
Descriptive	How have rice, vegetable, and fish prices moved over time (trends, seasonality, spikes)?
Predictive	Can we forecast the Metro Manila rice price 4 weeks ahead accurately enough to flag spikes early?
Prescriptive	Which shocks (typhoons, fuel, imports) are most associated with spikes, so the DA and traders know what to watch?

Dataset

- DA Bantay Presyo: daily and weekly market prices (the main target data)
- PSA price tables: official price statistics (for validation and context)
- Fuel prices: as a cost and transport shock
- Typhoon dates: as a supply disruption shock
- Import volumes: the third shock named in the question (the source still needs to be confirmed)

This has far more rows than the yearly yield tables, which suits time-series modeling better.

CRISP-DM Workflow

1. Business Understanding: Define the question, the users (DA and traders), and what counts as a “spike” (for example, a price rise above a set percentage within 4 weeks). Set success criteria such as forecast error (MAE/RMSE/MAPE) and spike detection recall.

2. Data Understanding: Check coverage, missing weeks or markets, and outliers. Explore trends, seasonality, and past spikes.
3. Data Preparation:
    - Clean and align daily and weekly prices to one weekly series.
    - Merge in fuel prices, typhoon flags, and import volumes.
    - Engineer lag features, rolling averages, and seasonal indicators.
    - Split by time (train on the past, test on the most recent period) to avoid leakage.
4. Modeling: Compare a naive baseline (last week's price), a classical model (ARIMA/SARIMA), and a machine learning model (Random Forest or Gradient Boosting with lag and shock features). Optionally add a spike-flagging layer on top of the forecast.
5. Evaluation:
    - Use time-series validation (walk-forward), not random splits.
    - Report forecast error against the baseline and how many real spikes were caught early.
    - Use feature importance to show which shocks matter most.
6. Deployment: Package the model to produce a 4-week-ahead forecast and spike alert that the DA and traders can read, such as a simple dashboard or report.